

Algebra Quiz 1: Rings

Time: 75 minutes

Total: 50 points

Name: TEJAS SAN

Date:

1. Prime ideals containing 2026

[10 points]

Describe all prime ideals $\mathfrak{p} \subset \mathbb{Z}[x]$ which contain 2026. Which of these prime ideals are maximal?

Hint: $2026 = 2 \cdot 1013$.

2. Automorphisms of a ring

[10 points]

Let R be a commutative ring.

- (a) Show that the set of ring isomorphisms $\psi : R \rightarrow R$ forms a group under composition. This group is denoted $\text{Aut}(R)$.
- (b) Show that every automorphism of $\mathbb{Z}[x]$ is of the form

$$x \mapsto \pm x + a, \quad a \in \mathbb{Z},$$

and deduce the structure of $\text{Aut}(\mathbb{Z}[x])$.

3. Idempotents modulo n

[15 points]

Let n be a positive integer. An element $e \in \mathbb{Z}/n\mathbb{Z}$ is called *idempotent* if $e^2 = e$.

- (a) Let $\omega(n)$ denote the number of distinct prime divisors of n . Show that $\mathbb{Z}/n\mathbb{Z}$ has exactly $2^{\omega(n)}$ idempotent elements.
- (b) Determine exactly when $\mathbb{Z}/n\mathbb{Z}$ has an idempotent different from both 0 and 1.
- (c) Determine the number of unital ring homomorphisms

$$\mathbb{Z}[x]/(x^2 - x) \longrightarrow \mathbb{Z}/n\mathbb{Z}.$$

4. Prime versus maximal

[5 points]

For each of the following ideals of $\mathbb{Z}[x]$, determine whether it is prime and whether it is maximal. Justify your answer by identifying the corresponding quotient ring:

$$(x), \quad (2, x), \quad (x^2 + 1).$$

5. A truncated polynomial ring

[10 points]

Let $n \geq 1$. How many ideals are there in the ring

$$A = \frac{\mathbb{C}[x]}{(x^n)}?$$

Which of these ideals are prime, and which are maximal?